

DR. DOC TUTORIALS

2018 ANATOMY EXAM

QUESTION-BY-QUESTION SOLUTIONS & STUDY MANUAL

A High-Precision, Fully Verified Review Guide with Comprehensive Explanations, Distracter Analyses, and Anatomical/Histological/Embryological Foundations.

Course Series:	Preclerkship Anatomy Unit (ITM)
Target Audience:	Medical Students (R)
Academic Reference:	Standardized Terminologia Anatomica & Gray's Anatomy
Branding Assurance:	Zero-Error Policy, Meticulously Double-Checked Key

SECTION I: HISTOLOGY & CELLULAR ANATOMY

Question 1: Regarding histology, choose the incorrect statement:

- A. It is the study of microscopic structures of the body
- B. It deals with cells and tissues regardless of the organs and systems they constitute
- C. It uses microscopes for examining structures at microscopic level
- D. It differs from gross anatomy in the size which details of the structure are revealed

Correct Answer: B. It deals with cells and tissues regardless of the organs and systems they constitute

Explanation: Histology is not simply the isolated study of cells and tissues in a vacuum. It is fundamentally concerned with understanding how these cellular components are organized and integrated into the specific organs and functional systems of the human body (often referred to as 'microscopic anatomy') [98]. The structure of any tissue or cell is intimately linked to the physiology and function of the organ it comprises. Thus, histology cannot study tissues regardless of their organ-level and system-level context.

Why other choices are incorrect:

- **A:** This is a true statement. Histology is defined as the scientific study of the microscopic anatomy of cells and tissues [98].
- **C:** This is a true statement. Because the structures of interest are microscopic, optical instruments (light or electron microscopes) are indispensable tools [98].
- **D:** This is a true statement. Gross anatomy focuses on structures visible to the naked eye, whereas histology resolves structures at the micrometer and nanometer scale [97, 98].

Question 2: The study of plasma membrane, mitochondria, endoplasmic reticula, and ribosomes is classified under which structural level of human body organization?

- A. Cellular level
- B. Tissue level
- C. Organismic level
- D. Chemical level

Correct Answer: A. Cellular level

Explanation: The cellular level is the level of structural organization where various molecules combine to form cells and their specialized functional subunits, known as organelles [104]. Subcellular components such as the plasma membrane, mitochondria, endoplasmic reticulum, and ribosomes are organelles that perform specific, life-sustaining functions within a cell, making their study a primary focus of the cellular level of organization.

Why other choices are incorrect:

- **B:** The tissue level consists of groups of similar cells and their surrounding extracellular matrix working together to perform a common function [104].
- **C:** The organismic level is the highest level of organization, representing the complete living individual [104].
- **D:** The chemical level is the most basic level of organization, involving individual atoms (such as carbon and hydrogen) and simple molecules (such as water or small ions) before they are assembled into complex cellular organelles [104].

Question 3: Which one of the following statements is NOT CORRECT describing tissues?

- A. Tissues are formed by groups of closely related cells and their product known as extracellular matrix
- B. Abundant microvilli which form prominent brush borders on luminal surface are the characteristic of epithelial tissues to have absorptive function
- C. Transitional epithelium is a type of epithelium whose appearance varies considerably depending on the degree of distension to which it is subjected
- D. Keratinized stratified squamous epithelium is normally for the body regions where the surface is dry or subject to mechanical abrasion
- E. Epithelial tissue is characterized by abundant extracellular matrix

Correct Answer: E. Epithelial tissue is characterized by abundant extracellular matrix

Explanation: Epithelial tissues are characterized by extremely high cellularity, with cells packed tightly together by cellular junctions. Consequently, they possess **very little extracellular matrix** (only a thin basal lamina separates them from underlying connective tissue) [11, 14]. In contrast, it is connective tissue that is characterized by an abundant extracellular matrix that widely separates the resident cells [14].

Why other choices are incorrect:

- **A:** This is a correct definition. All basic tissues are composed of specialized cells and their secreted extracellular products.
- **B:** This is correct. Microvilli on the apical surface of epithelial cells (such as in the small intestine or proximal renal tubules) form a brush border to drastically increase surface area for absorption [11].
- **C:** This is correct. Transitional epithelium (urothelium) lines urinary structures and transitions from a dome-shaped relaxed state to a flattened, thin stretched state as the organ distends with urine [11].
- **D:** This is correct. The keratinized layer of stratified squamous epithelium consists of dead, keratin-packed cells that provide an impermeable barrier protecting dry areas (like skin) from mechanical abrasion, water loss, and pathogens.

Question 4: Based on the number of nuclei, the cells indicated by the arrows are:

- A. Uninucleated
- B. Multinucleated
- C. Binucleated
- D. Anucleated

Correct Answer: C. Binucleated

Explanation: The histology slide referenced in the exam shows liver parenchymal cells (hepatocytes). While most hepatocytes contain a single nucleus (uninucleated), a significant proportion (up to 25-30%) are naturally ****binucleated**** (possessing exactly two nuclei). This occurs due to endomitosis or karyokinesis without cytokinesis. In basic medical training, hepatocytes are the classic histological example used to demonstrate binucleate cell pathology and normal cellular variations.

Why other choices are incorrect:

- **A:** Uninucleated cells contain only a single nucleus, which is the default state for the vast majority of human cells.
- **B:** Multinucleated cells contain three or more nuclei. Classic examples include skeletal muscle fibers, osteoclasts, and foreign-body giant cells.
- **D:** Anucleated cells lack a nucleus entirely. The prime example in humans is the mature erythrocyte (red blood cell), which expels its nucleus to maximize hemoglobin space.

Question 5: The part of a compound light microscope that is used to move the stage up and down during rough focusing is said to be:

- A. Fine adjustment knob
- B. Revolving nosepiece of turret
- C. Stage clip
- D. Iris diaphragm
- E. Coarse adjustment knob

Correct Answer: E. Coarse adjustment knob

Explanation: The **coarse adjustment knob** is designed to quickly and roughly focus the specimen. Turning this knob moves the microscope stage up or down rapidly over relatively large distances [12, 13]. It should only be used with low-power objectives (e.g., 4x or 10x) to avoid slamming the slide into high-power lenses.

Why other choices are incorrect:

- **A:** The fine adjustment knob is used for precise, high-power focusing, moving the stage in microscopic increments to bring out fine structural details [13].
- **B:** The revolving nosepiece (or turret) holds multiple objective lenses and rotates to switch between magnifications [13].
- **C:** The stage clip is a mechanical holder that keeps the glass slide firmly in place on the stage [13].
- **D:** The iris diaphragm regulates the diameter of the light beam passing up through the condenser, controlling contrast and illumination depth [13].

Question 6: The optical part of a compound light microscope that performs the functions of both magnification and resolution is:

- A. Condenser
- B. Illuminator
- C. Objective lens
- D. Eyepiece lens
- E. Both C and D

Correct Answer: C. Objective lens

Explanation: The **objective lens** is the single most critical optical component of a microscope because it performs both **primary magnification** and **resolution** (resolving power). The resolution of a light microscope depends entirely on the wavelength of light and the numerical aperture (NA) of the objective lens. The eyepiece lens (D) merely magnifies the virtual image produced by the objective, but **cannot add any new detail or resolution** (magnification without resolution is termed 'empty magnification'). Therefore, only the objective lens performs both functions.

Why other choices are incorrect:

- **A:** The condenser gathers and focuses the light rays from the illuminator onto the specimen but does not magnify the image [13].
- **B:** The illuminator is simply the light source (bulb or LED) that provides light [13].
- **D:** The eyepiece (ocular) lens provides secondary magnification of the image but does not contribute to the resolution.
- **E:** This is incorrect because the eyepiece lens (D) has no capability to resolve structures; resolving power is established solely by the objective lens.

Question 7: The part of a compound light microscope that is used to move the slide on the stage forward-backward and side-to-side to navigate through the specimen is:

- A. Coarse adjustment knob
- B. Revolving nosepiece of turret
- C. X-Y Adjustment
- D. Body tube
- E. Stage clip

Correct Answer: C. X-Y Adjustment

Explanation: The ****X-Y adjustment controls**** (also called mechanical stage controls or slide translation knobs) are connected to a mechanical stage. They are turned to move the slide clamp smoothly along the horizontal plane (Y-axis for forward-backward and X-axis for left-right), allowing precise navigation through the histological specimen [13].

Why other choices are incorrect:

- **A:** The coarse adjustment knob moves the stage vertically (up and down) for focusing, not horizontally [12, 13].
- **B:** The revolving nosepiece is rotated to change objective lenses, not to move slides [13].
- **D:** The body tube connects the eyepiece to the objective lenses [13].
- **E:** The stage clip holds the slide stationary relative to the mechanical stage but is not the mechanism used to translate/move it.

Question 8: All of the following subcellular structures are basophilic to common stains, EXCEPT:

- A. Mitochondria
- B. Nucleus
- C. Ribosomes
- D. RER
- E. Nucleolus

Correct Answer: A. Mitochondria

Explanation: Mitochondria are highly **acidophilic (eosinophilic)** structures. They contain a massive concentration of membranous proteins (including the electron transport chain complexes, ATP synthase, and Krebs cycle enzymes) which carry positive charges. These positive charges bind strongly to acidic, negatively charged dyes like **eosin**, staining them pink or red. In contrast, the other listed structures are rich in negatively charged nucleic acids (DNA in the nucleus, rRNA in ribosomes, RER, and nucleoli) which are **basophilic** because they bind basic dyes like **hematoxylin**.

Why other choices are incorrect:

- **B:** The nucleus is basophilic due to its high concentration of DNA and acidic histones.
- **C:** Ribosomes are highly basophilic because they are composed of ribosomal RNA (rRNA).
- **D:** The rough endoplasmic reticulum (RER) is highly basophilic because its membranes are studded with ribosomes containing rRNA.
- **E:** The nucleolus is basophilic because it is the active site of rRNA transcription and ribosomal subunit assembly.

Question 9: Tendons, ligaments, retinacula, cartilages and bones are chiefly composed of:

- A. Epithelial tissue
- B. Nervous tissues
- C. Muscle tissue
- D. Connective tissue

Correct Answer: D. Connective tissue

Explanation: These structures are all specialized or dense forms of **connective tissue** [14]. Connective tissues are characterized by abundant extracellular matrix containing fibrous proteins (such as collagen and elastic fibers) and ground substance that provide physical support, bind cells together, and transfer mechanical forces throughout the body.

Why other choices are incorrect:

- **A:** Epithelial tissue covers body surfaces and lines cavities; it lacks substantial extracellular matrix and does not form structural skeletal linkages [11].
- **B:** Nervous tissue consists of cells specialized for electrical signaling and integration (neurons and glia) [14].
- **C:** Muscle tissue consists of contractile cells specialized for producing force and movement.

Question 10: The basic tissue type that constitutes the brain and spinal cord is:

- A. Connective tissue
- B. Skeletal tissue
- C. Blood tissue
- D. Nervous tissue
- E. Epithelium

Correct Answer: D. Nervous tissue

Explanation: The central nervous system, including the brain and spinal cord, is primarily composed of **nervous tissue** [14, 50]. Nervous tissue is one of the four fundamental tissue types of the human body and consists of excitable signal-transmitting cells (neurons) and supporting cells (neuroglia) [50].

Why other choices are incorrect:

- **A:** Connective tissue forms the protective coverings of the CNS (meninges) but does not form the functional parenchyma of the brain or spinal cord [14].
- **B and C:** Skeletal and blood tissues are specialized sub-types of connective tissues, not separate primary tissue categories.
- **E:** Epithelium forms the lining of the brain ventricles (ependyma) but is only a minor cellular component of the overall CNS organ structure.

SECTION II: GROSS & SYSTEMIC ANATOMY

Question 1: What happens during dorsiflexion?

- A. The bottom of the foot moves toward the calf, decreasing the angle between those two surfaces, which leaves the toes pointing away from the body.
- B. The top of the foot moves toward the shin, decreasing the angle between those two surfaces, leaving the toes pointing toward the head.
- C. The foot twists to the left
- D. The foot twists to the right

Correct Answer: B. The top of the foot moves toward the shin, decreasing the angle between those two surfaces, leaving the toes pointing toward the head.

Explanation: Dorsiflexion is the term used to describe flexion of the foot at the ankle joint. This movement draws the **dorsum** (superior surface/top) of the foot superiorly toward the anterior aspect of the leg (shin), thereby decreasing the angle between them and leaving the toes pointed toward the head [113, 157]. This action occurs when walking uphill or when lifting the foot off the ground during the swing phase of gait.

Why other choices are incorrect:

- **A:** This describes **plantarflexion** (extension at the ankle), where the sole of the foot is pushed downward, pointing the toes away from the body (as when pressing a gas pedal) [113, 157].
- **C and D:** These options refer to **inversion** and **eversion**, which are lateral/medial twisting movements occurring at the subtalar and transverse tarsal joints, not the ankle joint [116, 159].

Question 2: Which motion best describes retraction?

- A. The scapulae move in a posterior/medial direction (posteromedial)
- B. The scapulae move in an anterolateral direction (anterolateral)
- C. The scapulae only move anteriorly
- D. The scapulae only move posteriorly

Correct Answer: A. The scapulae move in a posterior/medial direction (posteromedial)

Explanation: Retraction (also known as retrusion) of the shoulder girdle refers to the movement of the scapula on the thoracic wall in a ****posterior and medial**** direction [118, 161]. This action pulls the shoulders back and squeezes the scapulae closer to the vertebral column (midline). It is carried out primarily by the rhomboids and middle trapezius muscles.

Why other choices are incorrect:

- **B:** This describes ****protraction**** (protrusion), in which the scapulae slide anterolaterally around the rib cage, moving the shoulders forward (e.g., when reaching forward or throwing a punch) [118, 161].
- **C and D:** These are incomplete and inaccurate descriptions because scapular movement is not purely linear (anterior or posterior) but curvilinear, following the contours of the rounded thoracic cage.

Question 3: A patient involved in an automobile accident presents with a sharp object puncture of the middle of the sternum at about the level of the 4th or 5th costal cartilage. If the object also penetrated pericardium and heart wall, which heart chamber would most likely be damaged?

- A. Left atrium
- B. Left ventricle
- C. Right atrium
- D. Right ventricle

Correct Answer: D. Right ventricle

Explanation: The **right ventricle** forms the majority of the **anterior (sternocostal) surface** of the heart, lying directly behind the sternum and the 3rd to 5th costal cartilages. Because of this direct anatomical position, any anterior penetrating wound through the middle of the sternum at the 4th or 5th costal cartilage is highly likely to damage the right ventricle before any other chamber.

Why other choices are incorrect:

- **A:** The left atrium is situated on the posterior surface (base) of the heart and is shielded anteriorly by the ventricles and great vessels.
- **B:** The left ventricle forms the left border, apex, and diaphragmatic (inferior) surface of the heart, lying more laterally and posteriorly.
- **C:** The right atrium forms the right border of the heart and lies to the right of the sternal margin, making it less likely to be punctured by a midline sternal wound.

Question 4: Auscultation of the heart valves typically starts around the Angle of Louis (sternal angle). Which of the following sequences represents the correct order for locating the auscultation points as you move downward from the Angle of Louis?

- A. Pulmonic, Aortic, Mitral, Tricuspid
- B. Aortic, Pulmonic, Tricuspid, Mitral
- C. Aortic, Pulmonic, Mitral, Tricuspid
- D. Pulmonic, Aortic, Tricuspid

Correct Answer: B. Aortic, Pulmonic, Tricuspid, Mitral

Explanation: The sternal angle (Angle of Louis) lies at the level of the 2nd costal cartilage. Moving downward from this landmark, the clinical examiner auscultates the heart valves in a top-to-bottom sequence: 1. **Aortic point:** 2nd intercostal space, right sternal border. 2. **Pulmonic point:** 2nd intercostal space, left sternal border. 3. **Tricuspid point:** 4th/5th intercostal space, left sternal border. 4. **Mitral (Apical) point:** 5th intercostal space, left midclavicular line (the most inferior point, located near the heart apex). This sequence corresponds to the anatomical descending path: **Aortic -> Pulmonic -> Tricuspid -> Mitral**.

Why other choices are incorrect:

- **A, C, and D:** These options represent incorrect anatomical sequences that do not follow the orderly superior-to-inferior progression of clinical auscultation points.

Question 5: Which artery supplies the lower limb?

- A. Common iliac artery
- B. Internal iliac artery
- C. External iliac artery
- D. Common carotid artery

Correct Answer: C. External iliac artery

Explanation: The **external iliac artery** is the direct precursor to the main arterial supply of the lower limb. It descends through the pelvis and passes deep to the inguinal ligament to become the **femoral artery**, which supplies the thigh, leg, and foot [172].

Why other choices are incorrect:

- **A:** The common iliac artery is a pelvic trunk that bifurcates into the internal and external iliac arteries, but does not directly enter the limb.
- **B:** The internal iliac artery primarily supplies pelvic viscera, the perineum, and the gluteal region.
- **D:** The common carotid artery is located in the neck and supplies blood to the head and brain [101].

Question 6: Internal membrane of pericardium is:

- A. Pericardial cavity
- B. Parietal pericardium
- C. Serous pericardium
- D. Pericardial fluid

Correct Answer: C. Serous pericardium

Explanation: The pericardium is a fibroserous sac surrounding the heart. It consists of an outer tough fibrous pericardium and an inner, double-layered **serous pericardium** [184]. The serous pericardium is divided into a parietal layer (lining the inner aspect of the fibrous pericardium) and a visceral layer/epicardium (adhering directly to the heart muscle) [184]. Thus, the internal membrane of the pericardium is the serous pericardium.

Why other choices are incorrect:

- **A:** The pericardial cavity is the potential space between the parietal and visceral layers of the serous pericardium; it is a space, not a membrane [184].
- **B:** The parietal pericardium is merely one layer of the serous pericardium, whereas 'serous pericardium' describes the entire internal membrane system.
- **D:** Pericardial fluid is the lubricating serous fluid filling the pericardial cavity [184].

Question 7: The brain part that is responsible for thinking is:

- A. Cerebellum
- B. Hypothalamus
- C. Cerebrum
- D. Midbrain

Correct Answer: C. Cerebrum

Explanation: The **cerebrum** is the largest and most complex division of the human brain [104]. Its cerebral cortex is responsible for higher-order cognitive processing, abstract thinking, memory, intelligence, decision-making, planning, and the voluntary control of skeletal muscles [104].

Why other choices are incorrect:

- **A:** The cerebellum coordinates movement, posture, muscle tone, and motor learning, but does not process thoughts.
- **B:** The hypothalamus is the master control center for homeostasis, regulating temperature, hunger, thirst, circadian rhythms, and endocrine function.
- **D:** The midbrain is a brainstem region that coordinates visual and auditory reflexes.

Question 8: The part of the brain from which the spinal cord originates is:

- A. Pons
- B. Cerebellum
- C. Midbrain
- D. Medulla oblongata

Correct Answer: D. Medulla oblongata

Explanation: The **medulla oblongata** is the most inferior part of the brainstem. It sits in the posterior cranial fossa and extends inferiorly to pass through the foramen magnum of the occipital bone, at which point it becomes anatomically continuous with the **spinal cord**.

Why other choices are incorrect:

- **A:** The pons is the middle region of the brainstem, situated superior to the medulla and inferior to the midbrain.
- **B:** The cerebellum is located posterior to the pons and medulla.
- **C:** The midbrain is the most superior part of the brainstem, joining the diencephalon.

Question 9: Which one of the following is a cranial nerve that regulates heart rate?

- A. VII
- B. VIII
- C. IX
- D. X

Correct Answer: D. X

Explanation: Cranial nerve **X (the Vagus nerve)** provides the primary parasympathetic (visceral motor) innervation to the thoracic and abdominal organs [18, 101]. Presynaptic parasympathetic fibers of CN X synapse in cardiac ganglia located in the heart wall. When stimulated, these parasympathetic fibers secrete acetylcholine to **decrease the heart rate** and force of contraction.

Why other choices are incorrect:

- **A:** CN VII (Facial nerve) controls muscles of facial expression, taste on the anterior two-thirds of the tongue, and lacrimation/salivation.
- **B:** CN VIII (Vestibulocochlear nerve) is purely sensory, mediating hearing and balance.
- **C:** CN IX (Glossopharyngeal nerve) is involved in taste on the posterior third of the tongue, swallowing, salivary secretion (parotid gland), and monitoring blood pressure at the carotid sinus.

Question 10: The term proximal is best described as:

- A. Closer to the joint or extremity
- B. Further away from a joint or extremity
- C. Closer to the trunk of the body
- D. Further away from the trunk of the body

Correct Answer: C. Closer to the trunk of the body

Explanation: The directional term **proximal** describes a position that is **closer to the trunk** of the body or closer to the point of attachment/origin of a limb [111, 149]. For example, the shoulder is proximal to the elbow.

Why other choices are incorrect:

- **A and B:** These are non-standard descriptions because proximity is measured relative to the core trunk, not intermediate joints.
- **D:** This describes **distal**, which refers to a structure being further away from the trunk or the point of attachment [111, 149]. For example, the wrist is distal to the elbow.

Question 11: We call _____ a typical example of a branching cell.

- A. Red blood cell
- B. Muscle cell
- C. Neuron
- D. Egg cell

Correct Answer: C. Neuron

Explanation: A **neuron** is the classic biological example of a highly branched cell [19]. It has a central cell body (soma) from which numerous thin, branching cytoplasmic processes extend, including multiple dendritic branches (which receive incoming signals) and a long axon (which transmits signals) [19].

Why other choices are incorrect:

- **A:** Red blood cells (erythrocytes) are simple, unbranched, biconcave discs.
- **B:** Skeletal muscle cells are elongated, unbranched cylinders; cardiac muscle cells can branch slightly, but neurons are the absolute prime example of highly branching, tree-like structures.
- **D:** Egg cells (oocytes) are large, spherical, unbranched cells designed for fertilization.

Question 12: Which blood vessel supplies the majority of blood to the brain?

- A. Internal carotid artery
- B. External carotid artery
- C. Vertebral artery
- D. Subclavian artery

Correct Answer: A. Internal carotid artery

Explanation: The brain is supplied by two main arterial systems: the internal carotid system and the vertebrobasilar system. The **internal carotid arteries** supply approximately **80%** of the blood to the brain, supplying the anterior and middle portions of the cerebrum [19]. The vertebral arteries supply the remaining 20% (posterior cerebrum, brainstem, and cerebellum).

Why other choices are incorrect:

- **B:** The external carotid artery supplies superficial structures of the face, scalp, and neck, and does not supply the brain parenchymal tissue.
- **C:** The vertebral artery is a smaller system supplying only the posterior 20% of the brain.
- **D:** The subclavian artery is a major artery of the upper thorax/limb that gives rise to the vertebral artery, but does not directly supply the brain.

Question 13: The neuron that connects other neurons in the brain is known as:

- A. Sensory neuron
- B. Interneuron
- C. Motor neuron
- D. Receptor neuron

Correct Answer: B. Interneuron

Explanation: ****Interneurons**** (also known as association or integrative neurons) are located entirely within the central nervous system (brain and spinal cord) [20]. They form complex networks that act as intermediaries, connecting and coordinating communication between sensory (afferent) neurons and motor (efferent) neurons, which is the basis for integration and cognitive thinking [20].

Why other choices are incorrect:

- **A:** Sensory (afferent) neurons carry sensory information from peripheral receptors toward the central nervous system.
- **C:** Motor (efferent) neurons transmit motor commands from the CNS outward to effectors like muscles and glands.
- **D:** Receptor neurons are specialized sensory cells (or nerve endings) that detect external stimuli, not cells that link neurons inside the brain.

SECTION III: GENERAL EMBRYOLOGY

Question 1: Which of the following structures is unique compared to others in terms of embryonic origin?

- A. Epidermis
- B. Lens placode
- C. Neural crest cell
- D. Somite

Correct Answer: D. Somite

Explanation: The **somite** is derived from the **paraxial mesoderm** [20, 16]. In contrast, all the other listed options are of **ectodermal** origin: the epidermis (A) and lens placode (B) are derived from the **surface ectoderm**, and neural crest cells (C) are derived from the **neuroectoderm** during neurulation. Therefore, the somite is unique because it is the only mesodermal derivative listed.

Why other choices are incorrect:

- **A, B, and C:** These are all derived from the embryonic ectoderm, making them structurally related in origin.

Question 2: Concerning maturation of the female gamete (oogenesis), when do oogonia enter meiosis I and undergo DNA replication to form primary oocytes?

- A. At birth
- B. At puberty
- C. During fetal life
- D. With each ovarian cycle
- E. Following fertilization

Correct Answer: C. During fetal life

Explanation: Oogenesis begins **during fetal life** [20]. Oogonia multiply rapidly by mitosis. By the 5th month of gestation, they enter the first meiotic division (meiosis I) and undergo DNA replication to become **primary oocytes**. These primary oocytes are then arrested in the dictyate stage of prophase I, where they remain dormant until stimulated by hormones starting at puberty [20].

Why other choices are incorrect:

- **A and B:** No primary oocytes are formed after birth or at puberty. A female is born with her entire lifetime supply of primary oocytes already arrested in prophase I [20].
- **D:** With each ovarian cycle, a small cohort of pre-existing primary oocytes are stimulated to complete meiosis I, but they are not formed during the cycle.
- **E:** Fertilization triggers the completion of **meiosis II** in a secondary oocyte, not the formation of a primary oocyte.

Question 3: Assume that blastomeres were isolated from a blastula at the 4-cell stage. Each blastomere was cultured in vitro to the blastocyst stage and individually implanted into a pseudopregnant mother. What will happen?

- A. Birth of 1 baby
- B. Birth of 4 genetically different babies
- C. Birth of 4 genetically identical babies
- D. No birth

Correct Answer: C. Birth of 4 genetically identical babies

Explanation: Blastomeres at the very early stages of development (up to the 8-cell stage) are **totipotent**. Each cell retains the full genetic and developmental capacity to form a complete, normal embryo and extraembryonic membranes. Isolating the cells at the 4-cell stage and growing them independently results in the development of **four separate, genetically identical clones (quadruplets)** [21]. This is the cellular basis of identical twinning and cloning.

Why other choices are incorrect:

- **A:** This is incorrect because each of the four isolated cells has the potential to form a complete baby.
- **B:** Because all four blastomeres originated from a single fertilized zygote, they share identical DNA, meaning they cannot be genetically different.
- **D:** This is incorrect because totipotent cells can successfully implant and develop to full term under proper conditions.

Question 4: Brown-black lesions of skin due to sunburn (skin cancer / melanoma) most likely originate from which of the following embryonic derivatives?

- A. Endoderm
- B. Mesoderm
- C. Neural crest
- D. Neuroectoderm

Correct Answer: C. Neural crest

Explanation: Brown-black skin lesions from skin cancer are typically malignant **melanomas**, which arise from the malignant transformation of **melanocytes** (pigment-producing cells of the skin). Melanocytes are derived from **neural crest cells**, which delaminate from the neural folds during neurulation and migrate extensively through the dermis to populate the basale layer of the epidermis [21, 22].

Why other choices are incorrect:

- **A:** Endoderm forms the epithelial lining of the GI tract, respiratory tract, and associated glands [21].
- **B:** Mesoderm forms connective tissues, muscles, blood, bones, and the dermis of the skin, but not melanocytes [21].
- **D:** Neuroectoderm forms the neural tube (brain and spinal cord), but is not the direct progenitor of cutaneous melanocytes.

Question 5: If fertilization and implantation occurred, when would beta-hCG first be detectable in the maternal serum?

- A. 1st day after fertilization
- B. 3rd day after fertilization
- C. 8th day after fertilization
- D. 14th day after fertilization

Correct Answer: C. 8th day after fertilization

Explanation: Implantation begins around day 6 to 7 post-fertilization. The invading **syncytiotrophoblast** begins to secrete human chorionic gonadotropin (hCG). By **day 8** after fertilization (early in the second week), the syncytiotrophoblast has eroded maternal capillaries, allowing hCG to enter the maternal bloodstream in sufficient quantities to be detected by sensitive clinical serum essays [22].

Why other choices are incorrect:

- **A and B:** On days 1 and 3, the embryo is still a free-floating cleaving zygote/morula within the uterine tube or cavity, and has not yet established a connection to maternal blood [22].
- **D:** By day 14, hCG is abundant enough to be detected in maternal *urine* (basis of home pregnancy tests), but it is first detectable in the *serum* much earlier, on day 8.

Question 6: Treacher Collins syndrome arises from defective development of derivatives of which pharyngeal arch?

- A. First pharyngeal arch
- B. Second pharyngeal arch
- C. Third pharyngeal arch
- D. Fourth pharyngeal arch

Correct Answer: A. First pharyngeal arch

Explanation: Treacher Collins syndrome (mandibulofacial dysostosis) is an autosomal dominant disorder caused by mutations (e.g., in TCOF1) that impair ribosome biogenesis, leading to apoptosis of **neural crest cells**. This results in a massive deficiency of neural crest cells migrating into the **first pharyngeal arch** [22]. Consequently, first arch skeletal derivatives—such as the mandible, maxilla, and zygomatic bones—are hypoplastic, causing severe craniofacial malformations [22].

Why other choices are incorrect:

- **B:** The second pharyngeal arch (hyoid arch) forms the stapes, styloid process, lesser horn of hyoid, and facial muscles.
- **C:** The third pharyngeal arch forms the greater horn of the hyoid and stylopharyngeus muscle.
- **D:** The fourth and sixth pharyngeal arches form the laryngeal cartilages and muscles.

Question 7: hCG is produced by:

- A. Cytotrophoblast
- B. Syncytiotrophoblast
- C. Decidua basalis
- D. Mesoderm

Correct Answer: B. Syncytiotrophoblast

Explanation: The **syncytiotrophoblast** is the highly invasive, multinucleated outer layer of the trophoblast [23]. It synthesizes and secretes **hCG** directly into the maternal circulation to rescue the corpus luteum from degeneration, ensuring continued progesterone production and maintenance of the pregnancy [23].

Why other choices are incorrect:

- **A:** The cytotrophoblast is the inner, mitotically active cellular layer of the trophoblast that feeds cells into the syncytiotrophoblast but does not directly secrete hCG [24].
- **C:** The decidua basalis is the modified maternal endometrium at the site of implantation, not an embryonic hormone producer.
- **D:** The extraembryonic and intraembryonic mesoderm layers form structural tissues, not gonadotropic hormones.

Question 8: Which pharyngeal pouch expands and develops a solid, dorsal bulbar part and a shallow elongated ventral part?

- A. First pharyngeal pouch/arch
- B. Second pharyngeal pouch/arch
- C. Third pharyngeal pouch/arch
- D. Fourth pharyngeal pouch/arch

Correct Answer: C. Third pharyngeal pouch/arch

Explanation: The **third pharyngeal pouch** is characterized by its division into two distinct regions as it matures: a **dorsal bulbar part** and an **elongated ventral part**. The dorsal part differentiates into the **inferior parathyroid gland** (which migrates caudally with the thymus), while the ventral part fuses with its contralateral counterpart to form the **thymus**. Note: The exam choices substitute the word 'arch' for 'pouch', which is a minor typographical error; however, the anatomical association corresponds to the third pharyngeal system.

Why other choices are incorrect:

- **A:** The first pharyngeal pouch forms the tubotympanic recess, which becomes the tympanic cavity and auditory tube.
- **B:** The second pharyngeal pouch forms the tonsillar fossa and epithelial lining of the palatine tonsils.
- **D:** The fourth pharyngeal pouch forms a dorsal bulbar part (superior parathyroid gland) and a ventral part (ultimobranchial body, which contributes C-cells to the thyroid).

Question 9: Epiblast can form:

- A. Only endoderm
- B. Ectoderm and mesoderm
- C. Endoderm, ectoderm, and mesoderm
- D. Only ectoderm and endoderm

Correct Answer: C. Endoderm, ectoderm, and mesoderm

Explanation: Gastrulation is the landmark event of the 3rd week where the bilaminar embryonic disc is converted into a trilaminar embryonic disc [25]. All three definitive primary germ layers—**ectoderm, mesoderm, and endoderm**—are derived entirely from the **epiblast** [23]. Cells of the epiblast migrate through the primitive streak; some displace the hypoblast to form endoderm, others remain in the middle to form mesoderm, and the remaining epiblast cells form the ectoderm [23].

Why other choices are incorrect:

- **A, B, and D:** These are incorrect because they omit the epiblast's complete pluripotency in forming all three embryonic germ layers.

Question 10: Which structure is derived from the same embryonic primordium as the dorsal root ganglia?

- A. Kidney
- B. Gonads
- C. Pineal gland
- D. Liver

Correct Answer: C. Pineal gland

Explanation: The dorsal root ganglia (DRG) are derived from the **ectoderm** (specifically, **neural crest cells** which originate at the borders of the neural plate during neurulation). Among the choices listed, the **pineal gland** also develops directly from the **neuroectoderm** as a midline endodermal evagination of the diencephalon roof plate. Both are ectodermal, neural-system derivatives, sharing a close embryonic lineage compared to the mesodermal and endodermal organs.

Why other choices are incorrect:

- **A and B:** The kidneys and gonads are derived from the **intermediate mesoderm**, a completely different germ layer lineage [172].
- **D:** The liver is derived from the **endoderm** (specifically the hepatic diverticulum arising from the foregut endoderm).

Question 11: Which one of the following cells is multinucleated, has no mitotic fibers, and invades the endometrium?

- A. Cytotrophoblast
- B. Syncytiotrophoblast
- C. Hypoblast
- D. Epiblast

Correct Answer: B. Syncytiotrophoblast

Explanation: The **syncytiotrophoblast** is a multinucleated, cytoplasmic mass with no distinct cell boundaries (a syncytium) [24]. Because it is a multinucleate syncytium formed by cell-fusion, its nuclei do not undergo mitosis and **lack mitotic spindle fibers** [24]. Growth occurs entirely by the fusion of cells from the underlying cytotrophoblast. The syncytiotrophoblast secretes proteolytic enzymes that actively erode and **invade the maternal endometrium** to facilitate implantation [24].

Why other choices are incorrect:

- **A:** The cytotrophoblast is composed of individual, mononucleated cells that actively divide via mitosis (using mitotic fibers) to feed cells into the syncytiotrophoblast, but is not the primary invasive layer.
- **C and D:** The hypoblast and epiblast are mononucleated cellular layers of the bilaminar embryonic disc and do not invade the endometrium.

Question 12: Identify the mismatched pair between time and number of blastomeres during cleavage:

- A. 4-cell stage at 40 hr
- B. 2-cell stage at 30 hr
- C. 12-cell stage at 92 hr
- D. 16-cell stage at 96 hr

Correct Answer: C. 12-cell stage at 92 hr

Explanation: During early embryonic cleavage, cell divisions occur at predictable intervals. The standard timelines post-fertilization are: • 2-cell stage: ~30 hours (B is correct) • 4-cell stage: ~40 hours (A is correct) • 8-cell stage: ~72 hours (3 days) • 12-to-16-cell stage (Morula): ~96 hours (4 days) (D is correct). The designation of a '12-cell stage at 92 hr' (C) is anatomically non-standard and mismatched compared to these recognized landmarks.

Why other choices are incorrect:

- **A, B, and D:** These are highly accurate and standardized milestones of cleavage during week 1 of human development.

Question 13: In oogenesis, which event occurs immediately following completion of meiosis II?

- A. Degeneration of the zona pellucida
- B. Sperm penetration of the corona radiata
- C. Formation of a female pronucleus
- D. Appearance of the blastocyst
- E. Completion of cleavage

Correct Answer: C. Formation of a female pronucleus

Explanation: The secondary oocyte is arrested in metaphase of meiosis II at the time of ovulation. Meiosis II is completed ****only**** if a sperm penetrates the oocyte membrane. Once meiosis II is completed, the oocyte releases the second polar body, and its remaining haploid chromosomes decondense and form the ****female pronucleus**** [24]. This female pronucleus then aligns with the male pronucleus to complete fertilization.

Why other choices are incorrect:

- **A:** Degeneration of the zona pellucida ('hatching') occurs on day 5 to 6 of development, long after fertilization.
- **B:** Sperm penetration of the corona radiata is an early event of fertilization that precedes the completion of meiosis II.
- **D and E:** Cleavage and blastocyst formation occur several days after the completion of meiosis II.

Question 14: Which of the following events is not related to gastrulation?

- A. Formation of three germ layers
- B. Formation of primitive streak
- C. Cell migration and differentiation
- D. Formation of three body cavities
- E. Establishment of body axes

Correct Answer: D. Formation of three body cavities

Explanation: Gastrulation occurs in the 3rd week of development and is defined by the formation of the **three primary germ layers** (ectoderm, mesoderm, endoderm) from the epiblast, initiated by the appearance of the **primitive streak** and involving extensive **cell migration and differentiation** to establish the major **body axes** [25]. The **formation of the three body cavities** (pleural, peritoneal, and pericardial) is not part of gastrulation; it occurs later during the 4th week of development as a result of embryonic folding and the splitting of the lateral plate mesoderm [119, 173].

Why other choices are incorrect:

- **A, B, C, and E:** These are all core, classic developmental processes that define gastrulation during the third week of gestation [25].

Question 15: In the production of male gametes (spermatogenesis), which of the following cells remains dormant for about 12 years?

- A. Primary spermatocyte
- B. PGC (primordial germ cell)
- C. Secondary spermatocyte
- D. Spermatid
- E. Sperm

Correct Answer: B. PGC (primordial germ cell)

Explanation: Primordial germ cells (PGCs) migrate from the umbilical vesicle (yolk sac) to the genital ridges during the 5th week of development. In males, they are incorporated into sex cords where they differentiate into **gonocytes/prespermatogonia** and **remain completely dormant** during childhood. At **puberty** (around 12 years of age), these dormant cells resume activity, divide by mitosis to form definitive spermatogonia, and initiate spermatogenesis [25].

Why other choices are incorrect:

- **A, C, D, and E:** These cells are transient developmental stages formed during active, post-pubertal spermatogenesis and do not undergo long-term dormancy.

Question 16: A pregnant woman is at her 6-week prenatal appointment. The doctor informs her she is carrying twins. What is the most likely type of twin pregnancy?

- A. Diamniotic, dichorionic
- B. Diamniotic, monochorionic
- C. Monoamniotic, dichorionic
- D. Monoamniotic, monochorionic

Correct Answer: A. Diamniotic, dichorionic

Explanation: Statistically, ****diamniotic, dichorionic (DCDA)**** is the most common type of twin pregnancy. Approximately ****two-thirds (~70%)**** of all twins are ****dizygotic (fraternal)****, which always develop separate placentas and sacs, and are therefore always DCDA. Furthermore, even among ****monozygotic (identical)**** twins, about ****30%**** split early (at the morula stage), resulting in a DCDA configuration. Combined, DCDA represents the vast majority of all twin pregnancies.

Why other choices are incorrect:

- **B:** Diamniotic, monochorionic (MCDA) occurs in ~60-70% of ***monozygotic*** twins (when split occurs at the blastocyst stage), but is less common overall than dizygotic DCDA.
- **C:** This is biologically impossible; an embryo cannot share an amnion but have separate chorions.
- **D:** Monoamniotic, monochorionic (MCMA) is rare, occurring in only ~1-2% of monozygotic twins (when splitting occurs late, after amnion formation).

Question 17: Which of the following events is involved in cleavage of the zygote during week 1 of development?

- A. A series of meiotic divisions forming blastomeres
- B. Production of highly differentiated blastomeres
- C. An increased cytoplasmic content of blastomeres
- D. An increase in size of blastomeres
- E. A decrease in size of blastomeres

Correct Answer: E. A decrease in size of blastomeres

Explanation: Cleavage consists of a series of rapid mitotic divisions where the zygote splits into progressively smaller cells called blastomeres [26, 27]. Because the entire process occurs inside the rigid, non-expanding shell of the **zona pellucida**, there is no cellular growth phase between divisions. Consequently, with each round of division, the volume of cytoplasm is halved, resulting in a **progressive decrease in the size of the blastomeres** [27].

Why other choices are incorrect:

- **A:** Cleavage is a series of **mitotic** divisions, not meiotic.
- **B:** Blastomeres in early cleavage are undifferentiated and totipotent, not highly differentiated.
- **C and D:** Blastomeres decrease in size and retain the same total cytoplasmic volume divided among more cells.

Question 18: Which of the following structures produces progesterone late in pregnancy?

- A. Placenta
- B. Corpus luteum
- C. Syncytiotrophoblast
- D. Fetal adenohypophysis
- E. Maternal liver

Correct Answer: A. Placenta

Explanation: During the first trimester (up to weeks 8–10), the ovarian corpus luteum (B) is the primary source of progesterone. However, around the 10th week, the **placenta** takes over this role (the luteo-placental shift) [27]. In late pregnancy, the **placenta** (specifically its outer syncytiotrophoblast layer) synthesizes and releases massive quantities of progesterone to maintain uterine quiescence and prevent premature labor [27]. Both 'Placenta' and 'Syncytiotrophoblast' are accurate, but 'Placenta' is conventionally recognized as the primary endocrine organ of late pregnancy.

Why other choices are incorrect:

- **B:** The corpus luteum degenerates and plays a negligible role in hormone secretion in late pregnancy.
- **D and E:** The fetal pituitary and maternal liver do not produce progesterone.